Jacobian diagnostics for implicit ABBA

Implicit physical step and explicit application of its Jacobian

Adaptation to the GC2D problem with Hairer's symmetric projection

1 Purpose of the diagnostics

An implicit ABBA integration can be accompanied by a diagnostic that follows two objects:

$$z_n \in \mathbb{R}^2, \qquad \xi_n \in \mathbb{R}^2,$$

where z_n is the physical state and ξ_n is an infinitesimal perturbation or tangent vector. The state is computed using the complete implicit ABBA method,

$$z_{n+1} = \Psi_{h,t_n}(z_n),$$

whereas an optional observer updates the tangent vector by applying the ideal-root Jacobian of that same step:

$$\boxed{\xi_{n+1} = D\Psi_{h,t_n}(z_n)\,\xi_n.}$$

The responsibilities are deliberately separated:

- `ABBA2Implicit`, with the selected projection formulation, computes the trajectory and solves for the implicit multiplier μ_n;
- once μ_n has converged, a diagnostic observer can construct the tangent by means of a 2×2 linear system or an equivalent stage-increment recurrence.

Essential distinction

The Jacobian formula does not allow the physical step to be replaced by

$$z_{n+1} = D\Psi_{h,t_n}(z_n)z_n.$$

This equality is false for a general nonlinear map. The correct application is

$$\delta z_{n+1} = D\Psi_{h,t_n}(z_n)\,\delta z_n.$$

Therefore, z_{n+1} is first computed with the implicit method, and its Jacobian is then applied to ξ_n.

2 GC2D system and the small matrix w

The physical state is $z = (X, Y)^T$ and satisfies

$$\dot{z} = f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Let $s = h/2$. We introduce the *small* 2×2 matrix

$$\boxed{w(z,t) := s\,D_z f(z,t) = sJ_0 \nabla_z^2 \phi(z,t).}$$

For the GC2D vector field this matrix is explicitly

$$\boxed{w(z,t) = s\begin{pmatrix} -\phi_{XY}(z,t) & -\phi_{YY}(z,t) \\ \phi_{XX}(z,t) & \phi_{XY}(z,t) \end{pmatrix}.} \tag{1}$$

Since the Hessian of ϕ is symmetric,

$$w(z,t)^T J_0 + J_0 w(z,t) = 0.$$

The factor s is included in w so that the Jacobian of each elementary ABBA stage has the particularly simple form "identity plus one w block."

3 Implicit ABBA physical step

For a known state z_n and a provisional value $\mu \in \mathbb{R}^2$, the two copies are initialized as

$$u^{(0)} = z_n + \mu, \qquad v^{(0)} = z_n - \mu.$$

The four A–B–B–A stages are then applied:

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \tag{2a}$$
$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \tag{2b}$$
$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \tag{2c}$$
$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \tag{2d}$$

The multiplier for the step is the root $\mu_n = \mu_h(z_n)$ of the residual

$$\boxed{\mathcal{R}_n(z_n, \mu) := u^{(2)}(z_n, \mu) - v^{(2)}(z_n, \mu) + 2\mu = 0.} \tag{3}$$

At the exact root, the next physical state is

$$\boxed{z_{n+1} = u^{(2)} + \mu_n = v^{(2)} - \mu_n.} \tag{4}$$

The two expressions in (4) coincide because $\mathcal{R}_n(z_n, \mu_n) = 0$. With a finite nonlinear-solver tolerance, formulation 1 emits the neutral mean

$$\frac{1}{2}\left[(u^{(2)} + \mu_n) + (v^{(2)} - \mu_n)\right],$$

while formulation 2 emits the mean of its two simultaneous output unknowns. These representatives agree with (4) up to the nonlinear residual. Centered finite differences differentiate the emitted finite-tolerance map; the two analytic factorizations below evaluate the ideal-root tangent at the converged stages and can therefore differ from finite differences by the order of that residual.

4 Explicit calculation of the four w_i matrices

All quantities in this section are evaluated after the converged root μ_n has been obtained. The four small matrices are the matrix (1) evaluated at the four ABBA stage points:

$$w_1 := w(v^{(0)}, t_n) = s\begin{pmatrix} -\phi_{XY}(v^{(0)}, t_n) & -\phi_{YY}(v^{(0)}, t_n) \\ \phi_{XX}(v^{(0)}, t_n) & \phi_{XY}(v^{(0)}, t_n) \end{pmatrix}, \tag{5a}$$

$$w_2 := w(u^{(1)}, t_n) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_n) & -\phi_{YY}(u^{(1)}, t_n) \\ \phi_{XX}(u^{(1)}, t_n) & \phi_{XY}(u^{(1)}, t_n) \end{pmatrix}, \tag{5b}$$

$$w_3 := w(u^{(1)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_{n+1}) & -\phi_{YY}(u^{(1)}, t_{n+1}) \\ \phi_{XX}(u^{(1)}, t_{n+1}) & \phi_{XY}(u^{(1)}, t_{n+1}) \end{pmatrix}, \tag{5c}$$

$$w_4 := w(v^{(2)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(v^{(2)}, t_{n+1}) & -\phi_{YY}(v^{(2)}, t_{n+1}) \\ \phi_{XX}(v^{(2)}, t_{n+1}) & \phi_{XY}(v^{(2)}, t_{n+1}) \end{pmatrix}. \tag{5d}$$

For later use, define

$$\Sigma := w_2 + w_3. \tag{6}$$

5 Explicit calculation of the four stage Jacobians W_i

We now use a capital W_i for the 4×4 Jacobian of each elementary map in the duplicated variables (u, v). Each one is an identity matrix plus exactly one small w_i block.

For the first A stage,

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \qquad v^{(0)} = v^{(0)},$$

so

$$\boxed{W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_1 \\ 0 & 0 \end{pmatrix}.} \tag{7a}$$

If $w_1 = (w_{1,ij})$, this means explicitly

$$W_1 = \begin{pmatrix} 1 & 0 & w_{1,11} & w_{1,12} \\ 0 & 1 & w_{1,21} & w_{1,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For the first B stage,

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \qquad u^{(1)} = u^{(1)},$$

therefore

$$\boxed{W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_2 & 0 \end{pmatrix}.} \tag{7b}$$

Equivalently,

$$W_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{2,11} & w_{2,12} & 1 & 0 \\ w_{2,21} & w_{2,22} & 0 & 1 \end{pmatrix}.$$

For the second B stage,

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \qquad u^{(1)} = u^{(1)},$$

we obtain

$$\boxed{W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_3 & 0 \end{pmatrix}.} \tag{7c}$$

Equivalently,

$$W_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{3,11} & w_{3,12} & 1 & 0 \\ w_{3,21} & w_{3,22} & 0 & 1 \end{pmatrix}.$$

Finally, for the last A stage,

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}), \qquad v^{(2)} = v^{(2)},$$

which gives

$$\boxed{W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_4 \\ 0 & 0 \end{pmatrix}.} \tag{7d}$$

Equivalently,

$$W_4 = \begin{pmatrix} 1 & 0 & w_{4,11} & w_{4,12} \\ 0 & 1 & w_{4,21} & w_{4,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus the Jacobian of the complete four-stage ABBA map is the ordered product

$$\boxed{D\Phi_{h,t_n}^{\mathrm{ABBA}} = W_4 W_3 W_2 W_1.} \tag{8}$$

The order cannot be changed because the matrices w_i, and therefore the matrices W_i, do not commute in general.

6 Construction of the physical Jacobian

Multiplying the four matrices in (8), the Jacobian on the duplicated phase space can be written in 2×2 block form as

$$D\Phi_{h,t_n}^{\mathrm{ABBA}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = I_2 + w_4\Sigma, \tag{9a}$$
$$B = w_1 + w_4 + w_4\Sigma w_1, \tag{9b}$$
$$C = \Sigma, \tag{9c}$$
$$D = I_2 + \Sigma w_1. \tag{9d}$$

Notice that the factors s no longer appear separately in (9), because they are already contained inside every w_i.

To account for the implicit dependence $\mu_n = \mu_h(z_n)$, introduce the four 2×2 matrices

$$\begin{aligned}\mathcal{K} = 4I_2 &- (w_1 + w_2 + w_3 + w_4) \\ &+ (w_4\Sigma + \Sigma w_1) - w_4\Sigma w_1,\end{aligned} \tag{10}$$

$$\begin{aligned}\mathcal{L} = w_1 &- w_2 - w_3 + w_4 \\ &+ (w_4\Sigma - \Sigma w_1) + w_4\Sigma w_1,\end{aligned} \tag{11}$$

$$\mathcal{P} = I_2 + w_1 + w_4 + w_4\Sigma + w_4\Sigma w_1, \tag{12}$$

$$\mathcal{Q} = 2I_2 - w_1 - w_4 + w_4\Sigma - w_4\Sigma w_1. \tag{13}$$

Here,

$$\mathcal{K} = D_\mu \mathcal{R}_n, \qquad \mathcal{L} = D_z \mathcal{R}_n.$$

Differentiating the identity

$$\mathcal{R}_n(z, \mu_h(z)) = 0$$

with respect to z gives

$$\mathcal{L} + \mathcal{K} D\mu_h(z) = 0,$$

and, provided that $\mathcal{K}$ is invertible,

$$D\mu_h(z_n) = -\mathcal{K}^{-1}\mathcal{L}.$$

Consequently, the exact Jacobian of the implicit physical step is

$$\boxed{J_n := D\Psi_{h,t_n}(z_n) = \mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}.} \tag{14}$$

The matrix $\mathcal{K}$ in (10) is also the Jacobian of the residual used in Newton's method. However, $\mathcal{K}$ is not the physical Jacobian: the latter is J_n and also contains $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{L}$.

7 Applying the Jacobian without computing an inverse

In an implementation, $\mathcal{K}^{-1}$ should not be formed explicitly. To construct the entire Jacobian, solve the system with two right-hand sides:

$$\boxed{\mathcal{K}\mathcal{T} = \mathcal{L}, \qquad J_n = \mathcal{P} - \mathcal{Q}\mathcal{T}.} \tag{15}$$

Since $\mathcal{K} \in \mathbb{R}^{2\times 2}$, its factorization is inexpensive and can be reused for the two columns of $\mathcal{L}$.

7.1 Equivalent stage-increment factorization

The physical mean of the two converged copies can also be written as

$$z_{n+1} = z_n + \frac{h}{4}(k_1 + k_2 + k_3 + k_4),$$

where each k_i is the vector field evaluated at its traversed ABBA stage. Let

$$F_i = D_z f(\text{stage point}_i, \text{stage time}_i) = \frac{w_i}{s}, \qquad M_\mu = D\mu_h(z_n) = -\mathcal{K}^{-1}\mathcal{L}.$$

The total stage derivatives, including the dependence of the converged multiplier on z_n, obey

$$\begin{aligned}
Dk_1 &= F_1(I_2 - M_\mu),\\
Du^{(1)} &= I_2 + M_\mu + sDk_1,\\
Dk_2 &= F_2 Du^{(1)},\\
Dk_3 &= F_3 Du^{(1)},\\
Dv^{(2)} &= I_2 - M_\mu + s(Dk_2 + Dk_3),\\
Dk_4 &= F_4 Dv^{(2)}.
\end{aligned}$$

Consequently, a second diagnostic construction of the local physical Jacobian is

$$\boxed{J_n = I_2 + \frac{h}{4}(Dk_1 + Dk_2 + Dk_3 + Dk_4).}$$

This is an alternative computational factorization of (14), not a different numerical trajectory or a different mathematical tangent. Setting $M_\mu = 0$ would instead differentiate the explicitly averaged ABBA map.

If only a tangent vector ξ_n is to be propagated, it is not even necessary to form $\mathcal{T}$ or J_n. From (14),

$$J_n\xi_n = \mathcal{P}\xi_n - \mathcal{Q}\mathcal{K}^{-1}(\mathcal{L}\xi_n).$$

Therefore, compute successively

$$b_n = \mathcal{L}\xi_n, \qquad \mathcal{K}y_n = b_n,$$

and finally

$$\boxed{\xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n.} \tag{16}$$

The only implicit part of (16) is a 2×2 linear system; no new nonlinear equation appears.

7.2 Explicit formula for the 2×2 system

Let

$$\mathcal{K} = \begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}, \qquad b_n = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \qquad \Delta_{\mathcal{K}} = k_{11}k_{22} - k_{12}k_{21}.$$

If $\Delta_{\mathcal{K}} \neq 0$, the solution of $\mathcal{K}y_n = b_n$ is

$$\boxed{y_n = \frac{1}{\Delta_{\mathcal{K}}} \begin{pmatrix} k_{22}b_1 - k_{12}b_2 \\ -k_{21}b_1 + k_{11}b_2 \end{pmatrix}.} \tag{17}$$

This expression makes it possible to apply (16) without a general-purpose matrix solver.

8 One-step diagnostic workflow

Given an observed implicit ABBA step with t_n, z_n, ξ_n, h, and its converged stage snapshots, the diagnostic workflow is as follows:

1. Receive the multiplier μ_n and the required stage snapshots $u^{(0)}, v^{(0)}, u^{(1)}, v^{(2)}, u^{(2)}$.
2. Evaluate explicitly w_1, w_2, w_3, w_4 using (5), and form $\Sigma = w_2 + w_3$.
3. If the full duplicated Jacobian is needed, construct the four stage matrices W_1, W_2, W_3, W_4 from (7) and multiply them in the order $D\Phi^{\mathrm{ABBA}}_{h,t_n} = W_4W_3W_2W_1$.
4. Construct $\mathcal{K}$, $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{Q}$ using (10)–(13).
5. Select either the implicit-function factorization (15), the equivalent stage-increment recurrence, or centered finite differences of the emitted map.
6. Use the resulting J_n for symplecticity analysis and optional tangent propagation $\xi_{n+1} = J_n\xi_n$.

The observer does not replace or modify the already computed physical state z_{n+1}.

9 Propagation of the accumulated Jacobian

To obtain the derivative of the numerical solution with respect to the complete initial condition, propagate a matrix $M_n \in \mathbb{R}^{2\times 2}$ instead of a single vector. Initialize

$$M_0 = I_2$$

and, after each physical step, update it according to

$$\boxed{M_{n+1} = J_nM_n.} \tag{18}$$

Thus,

$$M_N = J_{N-1}J_{N-2}\cdots J_1J_0 = \frac{\partial z_N}{\partial z_0}.$$

The order of the products is important: the Jacobian of the most recent step always multiplies from the left.

The matrix update can also be performed without forming J_n. Solve

$$\mathcal{K}Y_n = \mathcal{L}M_n$$

for the two right-hand sides and set

$$\boxed{M_{n+1} = \mathcal{P}M_n - \mathcal{Q}Y_n.} \tag{19}$$

10 Symplecticity of the tangent propagation

If the projected ABBA step is solved exactly, its physical Jacobian satisfies

$$J_n^T J_0 J_n = J_0.$$

In two dimensions, this identity is equivalent to

$$\det J_n = 1.$$

Therefore, the application $\xi_{n+1} = J_n \xi_n$ is a symplectic linear transformation at each step. In addition, the accumulated product (18) is also symplectic:

$$M_N^T J_0 M_N = J_0, \qquad \det M_N = 1.$$

In numerical computations, it is useful to monitor

$$\varepsilon_{\mathrm{sp},n} := \left\| J_n^T J_0 J_n - J_0 \right\|, \qquad \varepsilon_{\det,n} := |\det J_n - 1|.$$

These quantities will not be exactly zero because of roundoff, the tolerance used to solve (3), and errors in the derivatives of ϕ.

11 Interpretation in terms of two nearby trajectories

Let $z_n^\varepsilon = z_n + \varepsilon \xi_n$, with ε small. Differentiability of the implicit map gives

$$\Psi_{h,t_n}(z_n^\varepsilon) = \Psi_{h,t_n}(z_n) + \varepsilon D\Psi_{h,t_n}(z_n)\xi_n + \mathcal{O}(\varepsilon^2).$$

Since $z_{n+1} = \Psi_{h,t_n}(z_n)$ and $\xi_{n+1} = D\Psi_{h,t_n}(z_n)\xi_n$, it follows that

$$z_{n+1}^\varepsilon = z_{n+1} + \varepsilon \xi_{n+1} + \mathcal{O}(\varepsilon^2).$$

Thus, the implicit integrator computes the reference trajectory while the optional diagnostic computes the linear evolution of an infinitesimal separation from it.

12 Implementation-ready summary

At each time step, perform the two updates

$$\boxed{\begin{array}{ll} z_{n+1} = \Psi_{h,t_n}(z_n) & \text{using ABBA and the implicit root } \mu_n, \\ \xi_{n+1} = \left(\mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}\right)\xi_n & \text{using a } 2 \times 2 \text{ linear system.} \end{array}}$$

The small matrices are

$$w_i = s\, D_z f(\text{stage point}_i, \text{stage time}_i),$$

and the four elementary 4×4 Jacobians are explicitly

$$\boxed{W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix}, \quad W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix}, \quad W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix}, \quad W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix}.}$$

Their ordered product is

$$D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1.$$

The recommended tangent update remains

$$b_n = \mathcal{L}\xi_n, \qquad \mathcal{K}y_n = b_n, \qquad \xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n.$$

The physical state is not obtained by multiplying z_n by the Jacobian. The Jacobian is applied to the tangent vector or, equivalently, to the accumulated Jacobian with respect to the initial condition.